Grey level Transformation.

A) Purpose of Grey level Transformation.

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Grey level transformation is used to improve the appearance of the image by changing the intensity values of pixels. It helps to enhance constract highlig-ht and makes objects easier to observe.

B) Negative Transformation

$$S = 255 - r.$$

Given $r =$ 50, 100, 150, 200.

Original Price	Transformed Pixel(s)
50	255 − 50 = 250
100	255 − 100 = 155
150	255 − 150 = 105
200	255 − 200 = 55

⇒ Dark pixels becomes bright.

⇒ Bight pixels becomes dark.

⇒ The image appears like a photographic negative making hidden more visible.

2) Histogram Equalization:

A) Histogram equalization is used to improve image contrast by spreading pixel intensities across the available intensity range making dark and bright. It helps to detect the objects, segmentation and visibility of a image.

Intensity	Frequency.
0	4
1	6
2	10
3	6.

Total Frequency $N = 4+6+10+6$

$= 26.$

PDF? formula?

Cumulative Frequency.

Intensity	Frequency	C.F	CDF
0	4	40	4/26 = 0
1	6	4+6=10	10/26 = 0
2	10	10+10=20	20/26=0
3	6	20+6=26.	26/26

c) Equalized Intensities

If a 4 level image (0 - 3)

$S = (L-1) \times CDF$

$S = (4-1) \times CDF$

$S = 3 \times CDF$

$3 \times 0.154 = 0.462 \approx 0$

$3 \times 0.355 = 1.065 \approx 1$

$3 \times 0.769 = 2.307 \approx 2$

$3 \times 1 = 3 = 3$

Intensity	CDF	Equalized intensity
0	0.154	0
1	0.355	1
2	0.769	2
3	1	3

3) Gaussian Smoothing

A) • Reduces noise smoothly without introducing sharp artifacts

• Preserves images edges better than a averaging filter

• Gives more weight to the center pixel that neighbouring pixels.

b) Given

Kernal $\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

Image $= \begin{bmatrix} 10 & 20 & 1 \\ 20 & 40 & \\ 10 & 20 & \end{bmatrix}$

$$= \frac{1}{16}\left[1(10) + 2(20) + 2(20) + 4(40) + 2(20) + 1(10\right.$$
$$\left. 2(20) + 1(10)\right]$$

$$= \frac{360}{16}$$

$$= 22.5 \approx 23.$$

The centre pixel for 40 to 23 reducing n
and smoothing the image while maintaining overall brightn

4) a) Edge Detection:

Edge Detection identifies boundaries where pixel intensity changes rapidly. It is used for object detection, edge detection and segmentation.

b) Horizontal Sobel Mask:

Mask : $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

Image: $\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$

$$= -1(10) + -2(10) + -1(10) + 0(20) + 0(20) + 0(20)$$
$$+ 1(30) + 2(30) + 1(30)$$

$$= -10 - 20 - 10 + 30 + 60 + 30$$

$$= -40 + 120$$

$$= 80$$

High gradient value (80) indicates a strong horizontal edge in the image.

A) Constrast Streetching.

Constrast stretching expands the intensity range to improve visibility and image quality.

b) Given $r = 50, 100, 150$

$$s = \frac{(r - r_{min})}{r_{max} - r_{min}} \times 255$$

where $r_{min} = 50$

$r_{max} = 150$

$r = 50$

$$s = \frac{(50 - 50) \times 255}{100} = 0$$

for $r = 75$

$$s = \frac{(75 - 50) \times 255}{100} = 63.7 \approx 64$$

For $r = 100$

$$s = \frac{(100 - 50) \times 255}{100} = 127.5 \approx 128$$

for $r = 125$

$$s = \frac{(125 - 50) \times 255}{100} = 191.25 \approx 191$$

For $r = 150$

$$s = \frac{(150-50) \times 255}{100} = 255$$

Original	New
50	0
75	64
100	128
125	191
150	255

∴ The intensity values now occupy the full range (0–255), producing a higher image with better visibility